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Data-driven Analysis of Chemical Short-range Order-Property Correlations in Multi-principal Element Alloys

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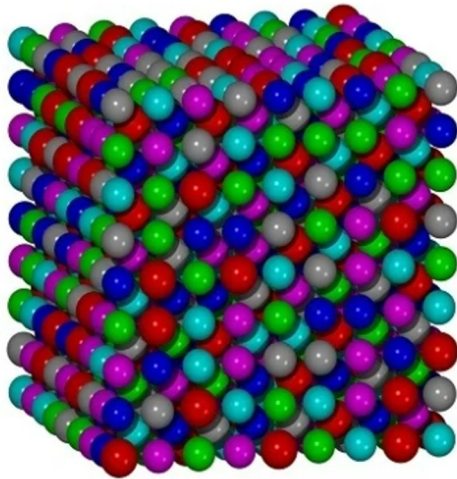
Joint Mathematics Meeting

January 9, 2025

Seattle, WA, USA



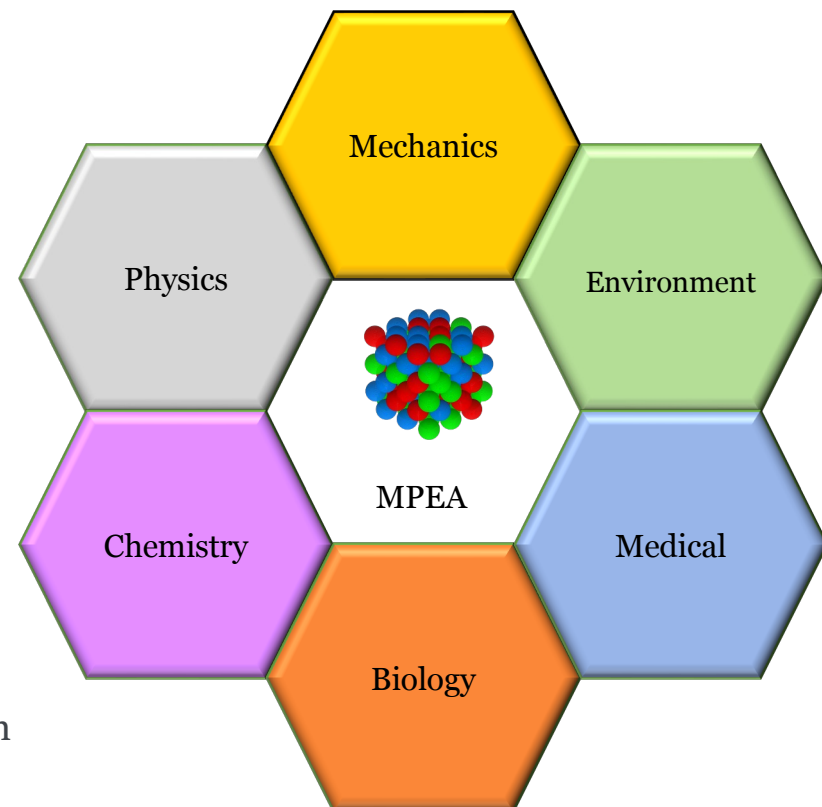
- Introduction
 - Multi-Principal Element Alloys (MPEAs)
 - Atomic Probe Tomography (APT)
 - Chemical Short-Range Order (CSRO)
- Methodology:
 - Equiatomic NiCoCr Alloy
 - Data processing and analysis
- Numerical Results and Discussions
- Conclusion



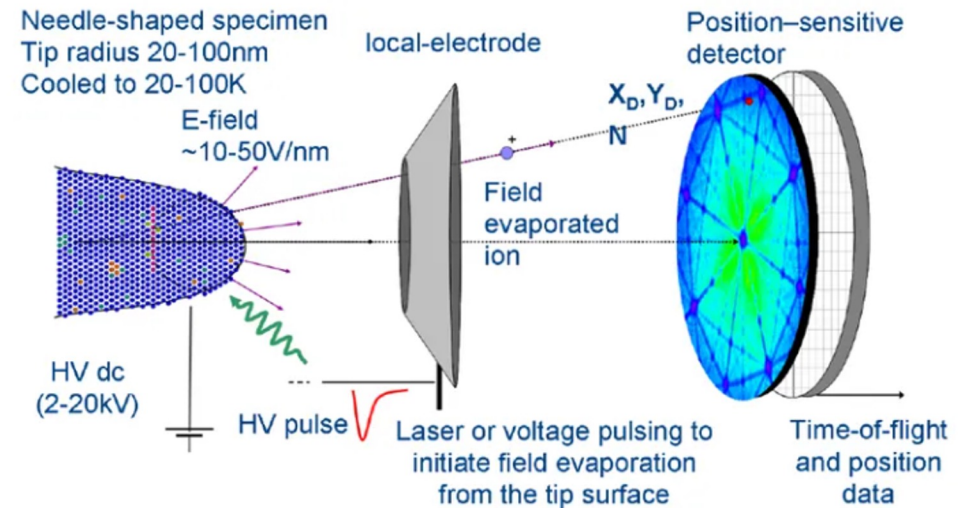
3D model of MPEA by Wang, S. 2013

- Near-equiatomic compositions of multiple elements
- High configurational entropy solid solution
- Complex microstructures with short-range ordering in practical

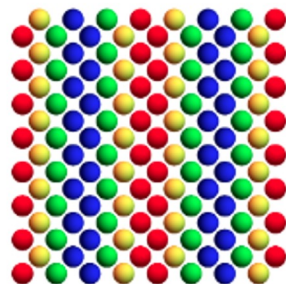
MPEA Applications



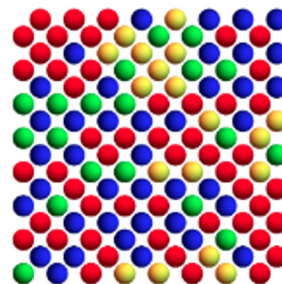
- Atomic Probe Tomography (APT) is a nanoscale characterization technique that maps the 3D atomic structure of materials by analyzing evaporated ions
- Limitations:
 - Data loss and spatial resolution artifacts.
 - Difficulty detecting CSROs due to random noise.
- Data-driven methods are needed to mitigate the limitations



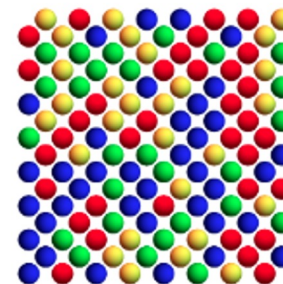
- Definition:
 - Local compositional ordering within a few atomic shells
- Characteristics:
 - Short-range and layered
 - Less complex than spatial SROs
- Goal: Employ ML techniques to identify CSRO in APT datasets



Long-range order



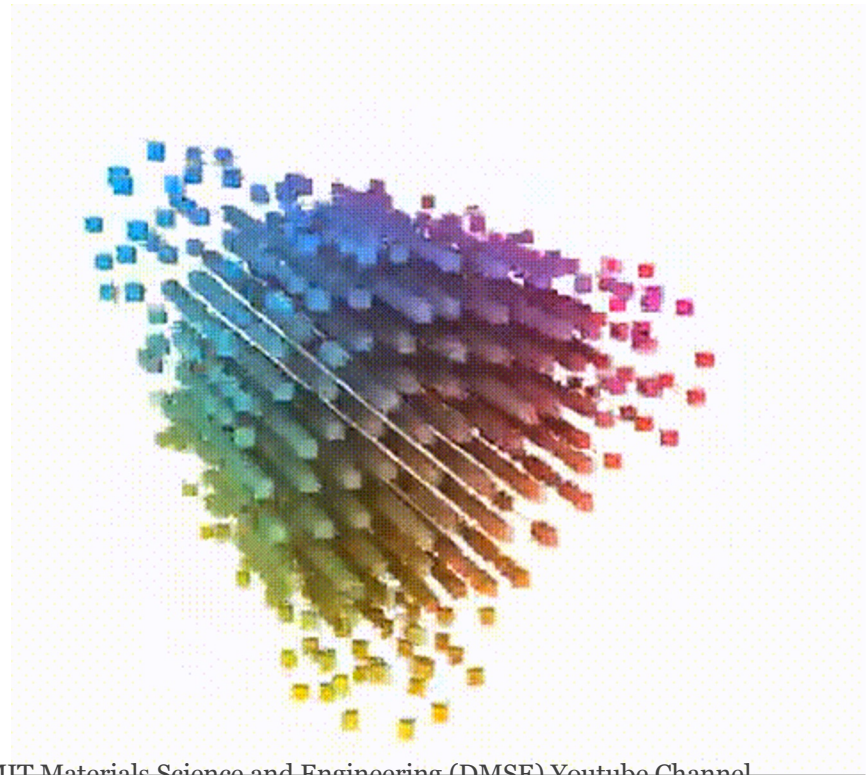
Short-range order



Random

Liu, X., Zhang, J., Eisenbach, M., & Wang, Y. (2019). Machine learning modeling of high entropy alloy: the role of short-range order. *arXiv: Materials Science*.

- Address the following challenges in chemical short-range order (CSRO) identification in MPEAs:
 - Analysis of complex microstructures in MPEAs
 - Limitations of Atomic Probe Tomography (APT) in experimental data reconstruction and resolution
 - Need for quantitative metrics to link CSRO to various properties
- Enable rational design of advanced alloys with novel properties by understanding CSRO-property relationships

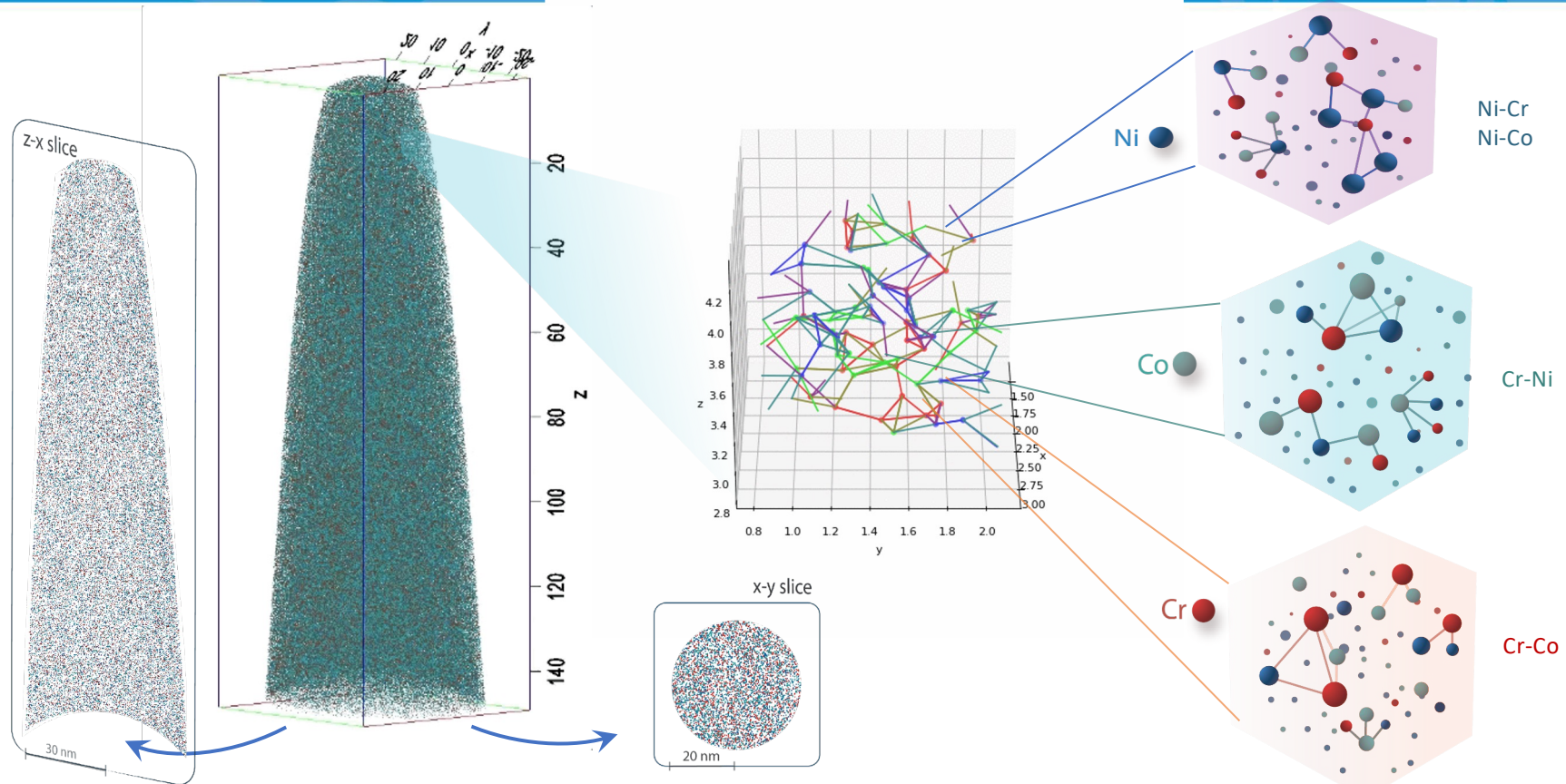


MIT Materials Science and Engineering (DMSE) Youtube Channel



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ML-aided Pattern Detection in NiCoCr Alloy

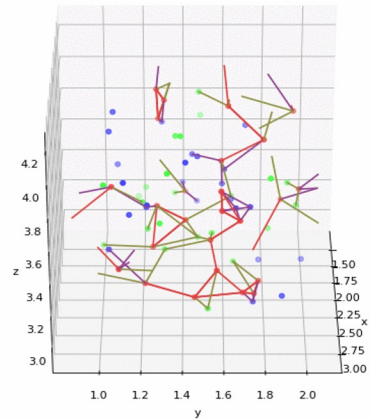


Shan Cecilia Cao, R. Feng, Y. Zhang, H.Y. Sun, Lei Lu, Jian Lu, Z. Wang, C.T. Liu, Robert O. Ritchie, *Discovering Short Chemical Preference in High-entropy Alloys by Real-time Machine Learning*, 2019

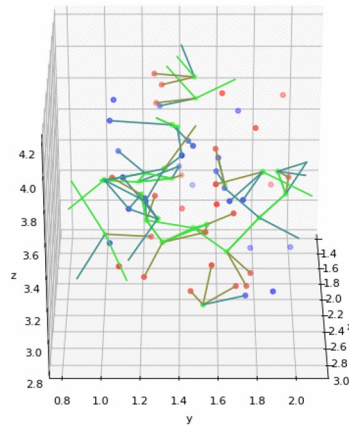


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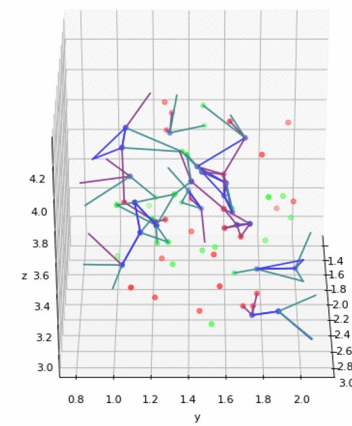
ML-aided Pattern Detection in NiCoCr Alloy



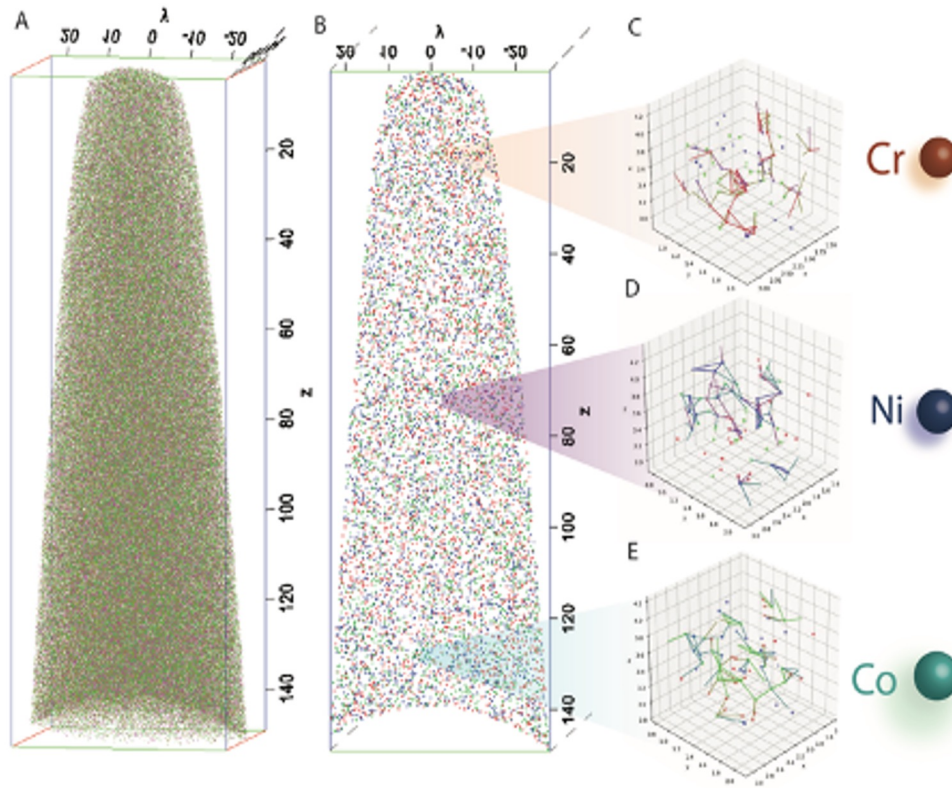
Cr is the central atom
The nearest four atoms of each Cr in the area.



Co is the central atom
The nearest four atoms of each Co in the area.



Ni is the central atom
The nearest four atoms of each Ni in the area.

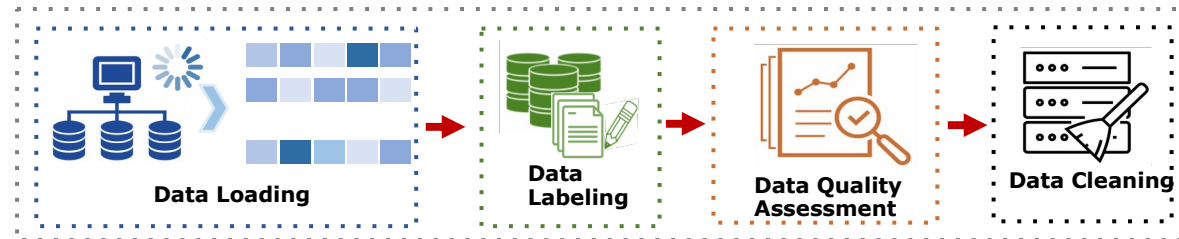


- Preparation:
 - Remove outliers and noise
 - Assign atomic labels to 3D point map
- Nearest neighbor-based metrics:
 - Focusing on constitutional information
 - Mitigate intrinsic limitation of APT datasets
- Quantitative CSRO indicator
 - To check correlation to properties
 - To lead new material/ preparation method design

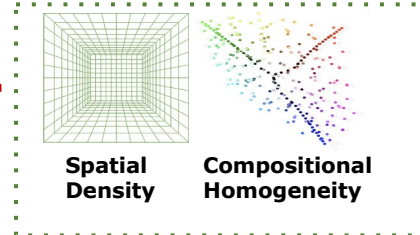
Data Pipeline Flow Chart



Data Preparation



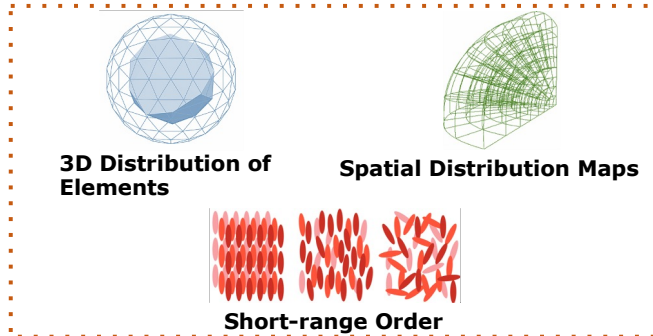
Exploratory Analysis



Data Pipeline Development



Visualizations



- Load data files and data exploration
- Assign element labels

Example rows of the dataset

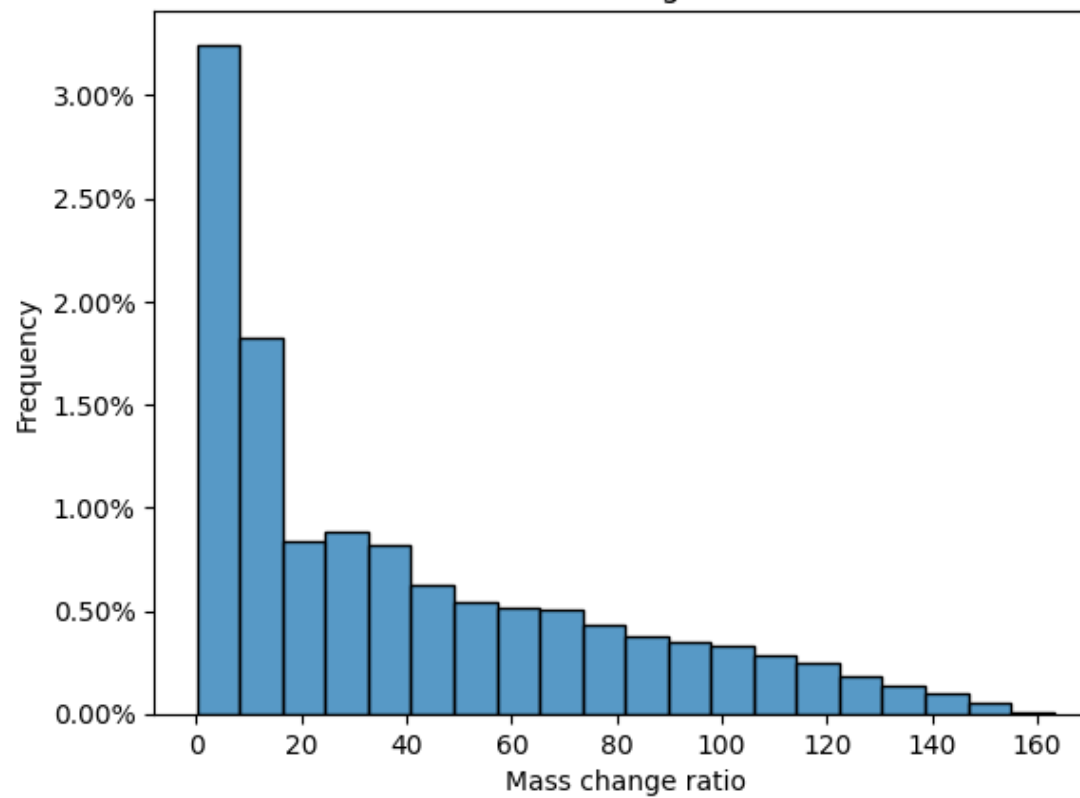
	Xnm	Ynm	Znm	Mass	Element
0	-7.306091	-5.917572	2.748160	29.053513	Ni
1	0.538717	1.667120	0.088205	19.736074	Unknown
2	-2.756541	-9.206660	2.883611	28.945791	Ni
3	-5.937352	4.714521	1.732376	25.987719	Cr
4	-4.844893	4.721588	1.364333	25.996983	Cr
5	-3.088373	4.203208	0.797689	25.977236	Cr
6	-4.348228	4.870821	1.267384	28.980291	Ni
7	-3.212724	5.102966	1.074952	29.453442	Co
8	-5.038929	5.127590	1.549493	29.997074	Ni
9	-0.890082	-2.170647	0.158911	29.971703	Ni



Noise Distribution

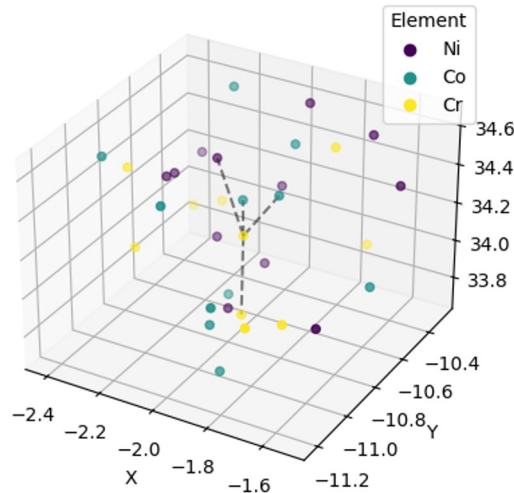


Distribution of Mass/Charge Values of Outliers

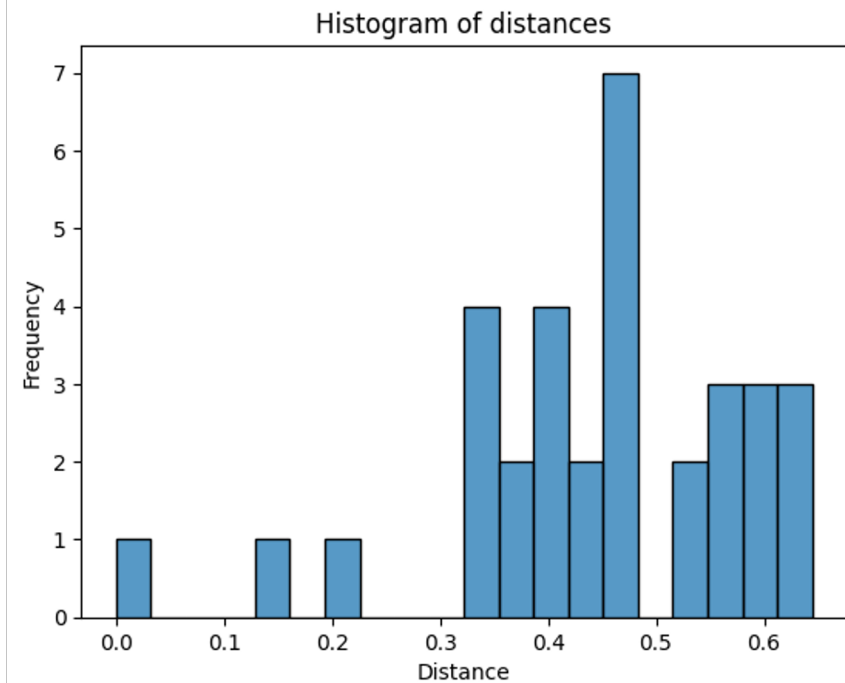
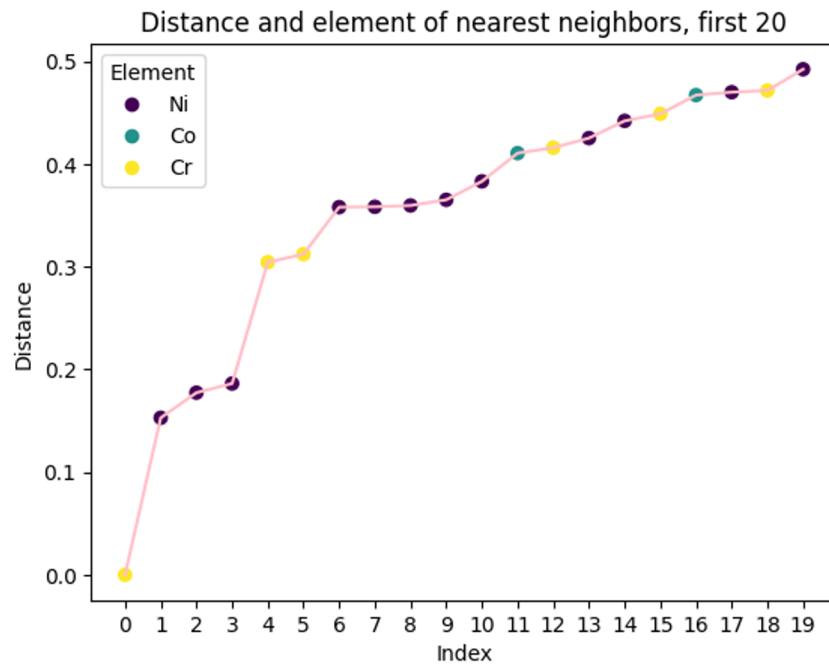


Composition analysis: Close to equiatomic 1:1:1

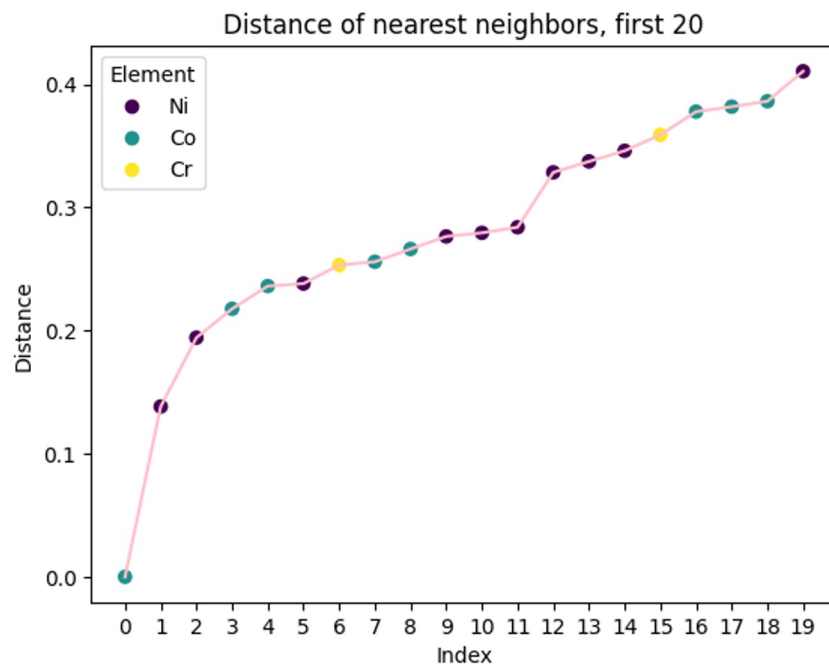
```
Element
Ni      33.608661
Co      33.501881
Cr      32.889458
Name: proportion, dtype: float64
```



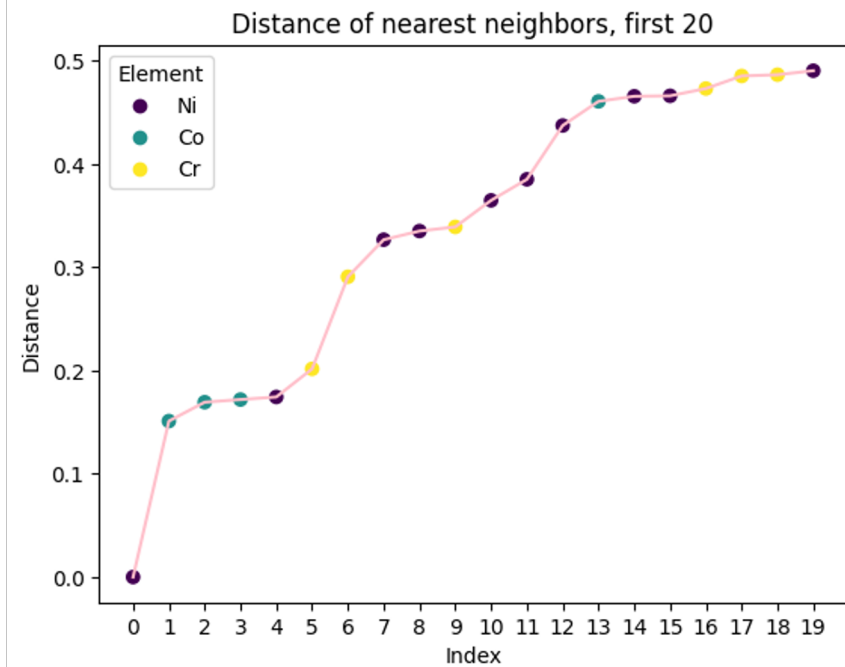
- Spatial Density Analysis
 - Number of atoms per cubic nanometer
- Example:
 - 319861 atoms are sampled in a cube with 20 nm edge length
 - 39.9826 atoms per cubic nanometers.
- Thus, it is safe to sample in a cube with 2 nm edge length to search for at most 50 neighbors.



Chromium Center

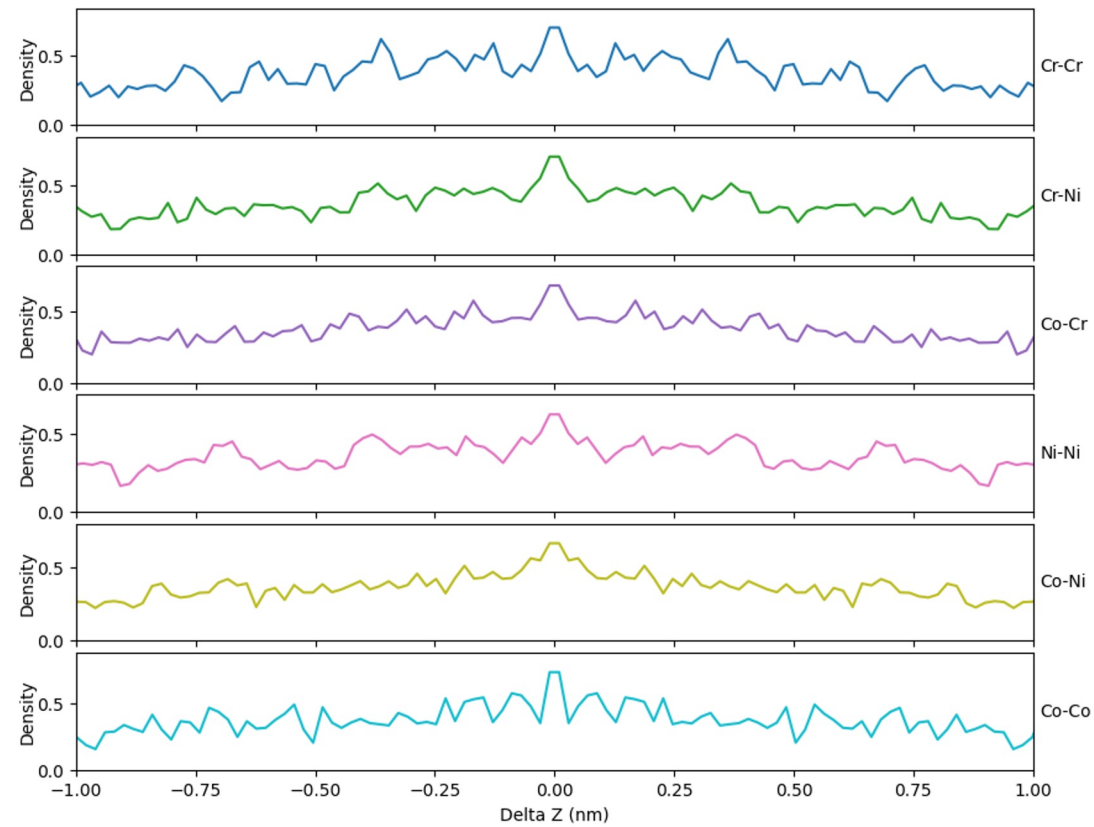


Cobalt Center

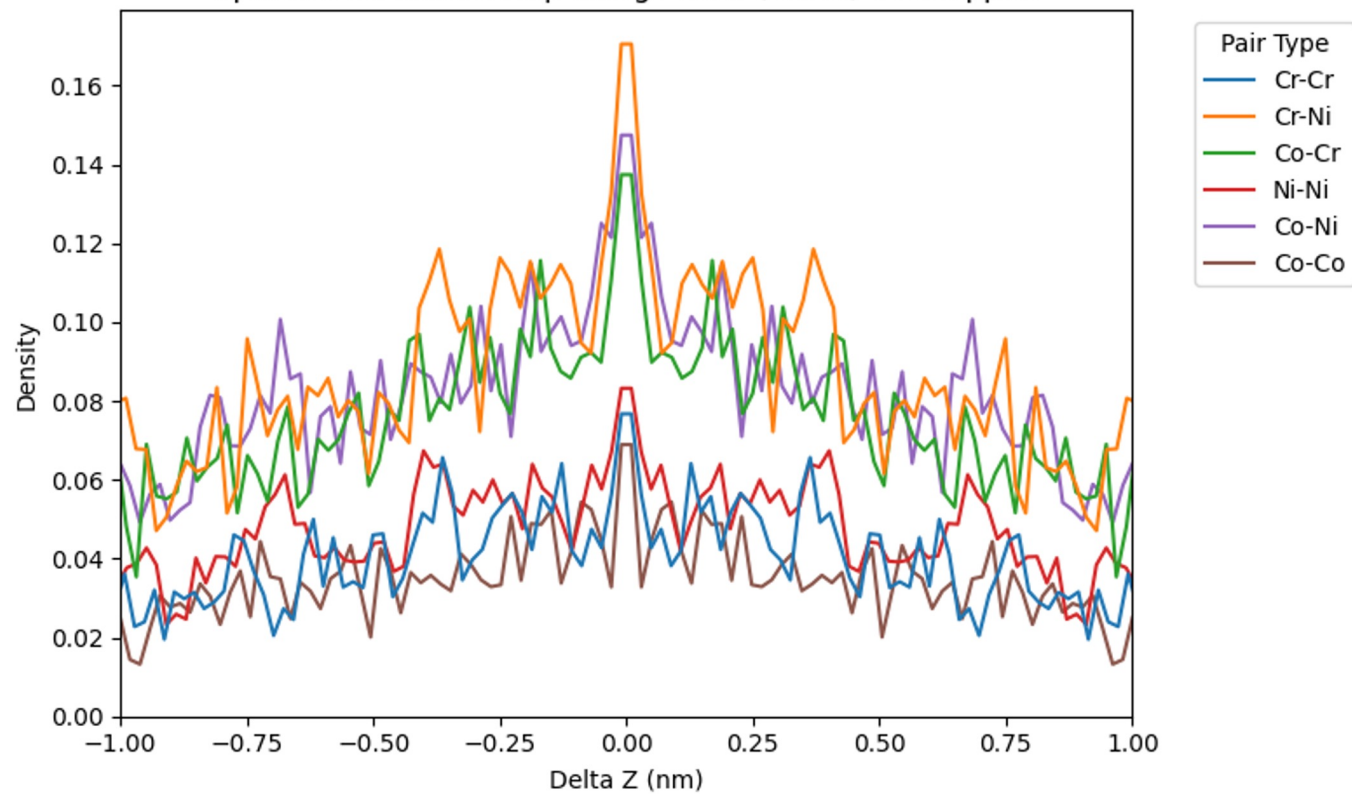


Nickel Center

Spatial Distribution Map along Z axis (zSDM)



Spatial Distribution Map along Z axis (zSDM) Overlapped



- Understand chemical short-range order (SRO) in multi-principal element alloys (MPEAs) is essential.
- Addresses challenges of complex microstructures, APT data limitations, and the need for quantitative CSRO-property metrics.
- Development of robust analytical methods and tools is crucial.
- Enabling precise establishment of CSRO-property relationships is essential.



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